

Generation and Modulation of Second-order Nonlinearity in the Microfibre

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Abstract We demonstrate optical fibre schemes assisted with low-dimensional materials to implement second-order nonlinear processes. The functionalized fibre devices have achieved tunable and broadband second harmonic and sum-frequency generation under low peak-power CW lasers, facilitating applications in nonlinear fibre optics. ©2026 Yuxin Ma and Biqiang Jiang

Introduction

Second-order nonlinear optical effects hold considerable potential for various fields such as optical frequency converters, advanced lasers, biological imaging, and microscopy technologies [1-4]. They are also eagerly anticipated for applications in optical silica fibres but are extremely weak because of the centrosymmetric nature of the material. However, **the prior successes of second-order nonlinearity in fibre mainly rely on complex poling techniques [5-7], requiring challenging and harsh fabrication conditions, which are not conducive to their development.**

With the emergence of nanomaterials, numerous nonlinear materials offer excellent performance in silicon-based photonic devices [8,9]. **Inspirationally, the material-integrated fibre platforms provide a solution for the excitation of second-order nonlinear effects [10-14].** Owing to the high-transmission and low-dispersion modes of fibre waveguides, the second harmonic (SHG) and sum-frequency generation (SFG) processes are expected to be effectively enhanced by adopting the strong and long-distance interactions between the localized light field and integrated nonlinear materials.

In this work, we present an all-fibre strategy that achieves the tunable second-order frequency conversions. The microfibre allows precise waveguide dispersion engineering to satisfy effective interaction with integrated gallium selenide (GaSe). Switch to multiple pumps, we have successfully transferred both SHG and SFG to broadband excitations, extending about 180-nm SHG and 31.7-nm SFG. Moreover, when the device is transformed into a microfibre knot resonator (MKR), the resonant SHG is enhanced and supported by polarization-dependent control by split modes in the twisted structure. The proposed scheme provides a practical route to controllable, enhanced second-order frequency mixing in all-fibre architectures, with immediate implications for nonlinear parametric control, widely tunable broadband sources, optical fibre

communications, and sensing.

Multiple and broadband frequency mixing processes

The layered GaSe-coated microfibre is schematically illustrated in Fig. 1(a), along with the crystal structure of ϵ -GaSe employed in the following experiment. The tapered region of microfibre possesses a diameter of around 3 μm . The used ϵ -GaSe exhibits a high second-order nonlinear susceptibility and keeps the integration length of around 102 μm . The simulation of waveguide dispersion of possible modes shows that, the precise phase-matching between $\text{HE}_{11}(\omega)$ and $\text{HE}_{31}(2\omega)/\text{EH}_{11}(2\omega)$ modes is realized in the microfibre with a diameter of 3.18 μm , as they hold the same effective refractive index. Their minor phase mismatch over a wide wavelength range lays a solid foundation for frequency conversions across a wide spectrum.

As the four CW lasers were combined to jointly pump the GaSe-coated microfibre, four SHG signals and six SFG signals were excited, which are shown in Fig. 1(b). To extremely explore the bandwidth of the produced SHG signal, an SC light source with a near 430 nm bandwidth was tentatively employed to provide the pump light. Figure 1(c) displays the excited ultra-broadband SHG (ultra-BSHG) continuum under the pump power of 6.5 mW. The bandwidth of BSHG is effectively extended to 179.7 nm (more than 6.3 times) compared with our previous work [15].

In addition, two SLED light sources, centering at 1550 nm (SLED₁, 25.5 mW, 102.7 nm bandwidth) and 1300 nm (SLED₂, 12.9 mW, 83.9 nm bandwidth), were organized to extremely investigate the bandwidth of the generated SFG signal. The typical spectrum obtained through measurement is shown in Fig. 1(d). Respectively, the left and right peaks correspond to the SHG signals of the two SLEDs, while the middle peak is the SFG signal excited synchronously by both SLEDs. The overall spectrum is fitted by the sum

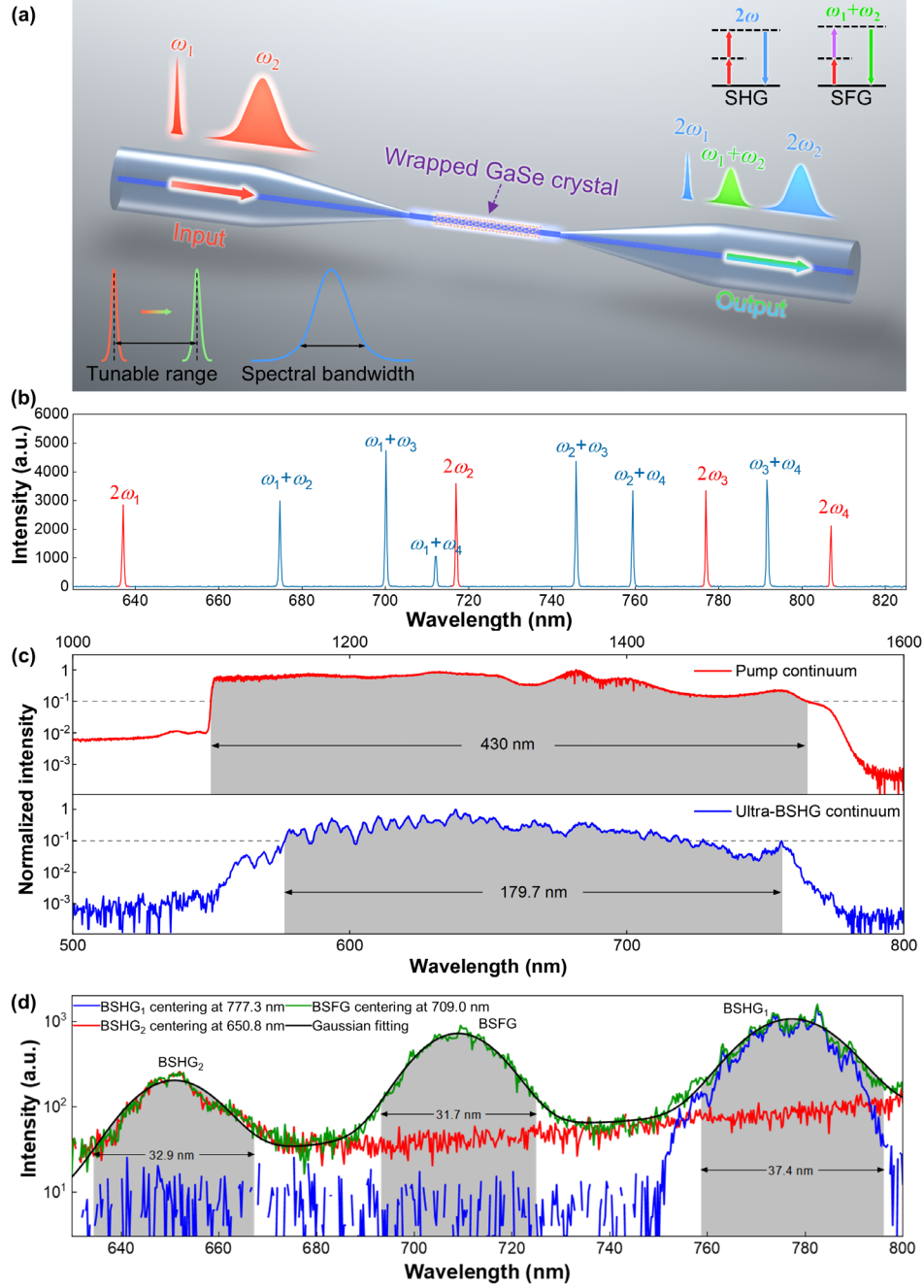


Fig. 1: Multiple and broadband second-order nonlinear processes.

of three Gaussian functions. Thus, the 10-dB bandwidths of the BSHG were 32.9 nm and 37.4 nm, and the value of BSFG is about 31.7 nm. Obviously, the GaSe-integrated hybrid microfiber can greatly serve as a developed platform for broadband optical frequency conversion.

Polarization-Controlled SHG Enhancement

Further, the microfiber was fabricated to the GaSe-MKR structure shown in Fig. 2(a), where the inherent strong $\chi^{(2)}$ of GaSe, coupled with the high-Q resonant modes supported by the MKR, greatly enhances the second-order nonlinear

interaction. The adopted MKR exhibits major and minor axes measuring 940 μm and 830 μm , respectively. The GaSe integration demonstrates uniform surface coverage across 67.6 μm of the loop region, which is shown in the inset of Fig. 2(b). When pumped with near 1550 nm laser, the excitation of SHG at resonant wavelengths realizes an obvious enhancement compared to GaSe-microfiber integration. Figure 2(b) displays the intensity of their SHG spectrum.

Additionally, the pump wavelength was scanned from 1548.4 nm to 1552.1 nm, the corresponding SHG spectra (solid lines)

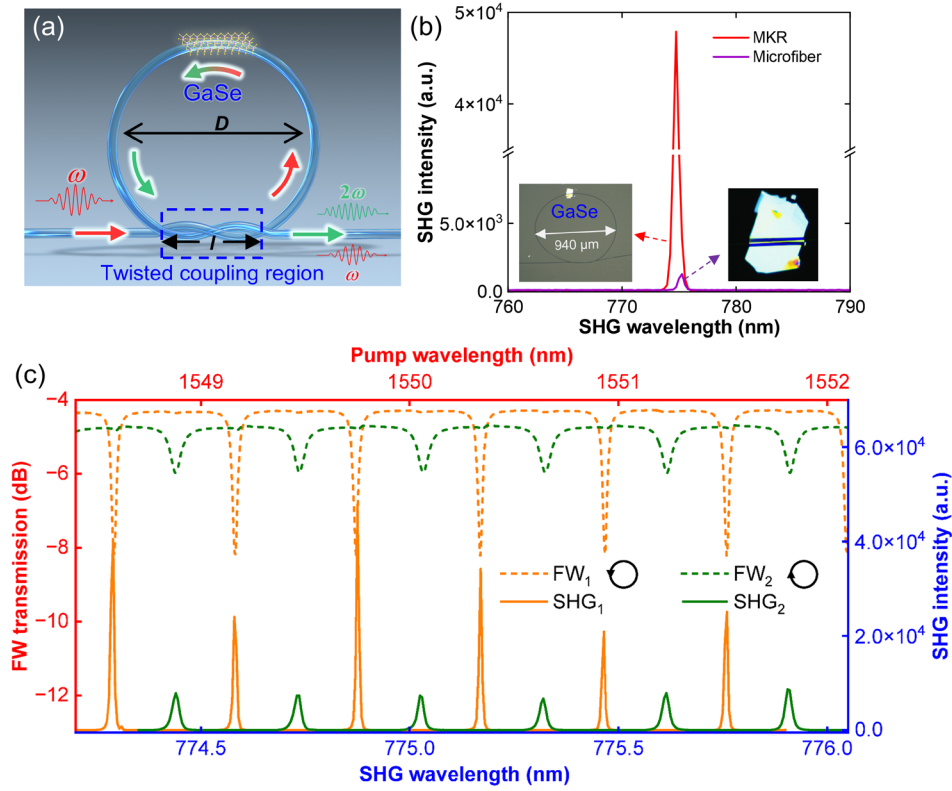


Fig. 2: Polarization-controlled come-like SHG signals in hybrid MKR

demonstrate a significant enhancement at the resonant wavelengths and reveal the critical polarization-dependent behaviour in Fig. 2(c). When the polarization state of the pump light is adjusted, the corresponding positions of comb-like SHG resonance enhancement shift accordingly, achieving selective wavelength modulation. Compared to a conventional micro-ring resonator, the twisted MKR with an interleaved coupling architecture provides enhanced resonance stability, uniform multi-frequency conversion efficiency, and improved optical field overlap, all of which facilitate more efficient parametric processes.

Conclusion

In summary, we have demonstrated a practical all-fibre approach to broadband optical frequency conversion based on second-order nonlinearity in a GaSe-coated hybrid microfiber. Benefitting from the large interaction length, ultrahigh $\chi^{(2)}$ response, and long-range crystalline order of layered GaSe crystal, both narrowband and broadband SHG and SFG processes can be readily realized, especially the 180-nm ultra-BSHG and 31.7-nm BSFG. Furthermore, the developed hybrid MKR achieved an excellent improvement over the conventional microfiber structure. The spectral engineering of the comb-like fundamental frequency resonances ensured stable SHG with discrete spectral uniformity. Our

scheme offers a versatile, scalable platform to extend the operational bandwidth and optional intensity of fibre nonlinear optics, with direct implications for tunable broadband sources, reconfigurable photonic systems, fibre communications, and sensing.

Acknowledgements

We are grateful for financial supports from National Natural Science Foundation of China (Nos. 62375223, 62322510); Natural Science Basic Research Plan in Shaanxi Province of China (2025JC-QYXQ-030)

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